Tapas Ranjan Rath

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Research Profile

Computational cognitive scientist studying how comparison structure and task context shape choice and preference construction. Doctoral research on the attraction effect is complemented by separate eye-tracking research on judgments of AI-generated video and a developing program on agency, social consequences, and human control in AI-assisted action. Other work includes topological analysis of EEG signals.

Education

Indian Institute of Technology Kanpur 2019–present

Ph.D., Cognitive Science *Thesis submitted: 8 July 2026*

- Defense expected in early November 2026.
- Thesis: *When and Why Context Matters: Establishing Boundary Conditions and Mechanisms of the Attraction Effect in Decision-Making.*
- Supervisors: Prof. Narayanan Srinivasan and Prof. Nisheeth Srivastava.

SRM University, Chennai 2008–2012

B.Tech., Electronics and Instrumentation Engineering

Publications and Manuscripts

Journal and Conference Publications

- **Rath, T.**, Srinivasan, N., & Srivastava, N. (2025). The attraction effect in perceptual decision-making: A case of dominance asymmetry. *Frontiers in Psychology, 16*, 1661748. [Article]
- **Rath, T.**, Srivastava, N., & Srinivasan, N. (2026). A self-directed expanded judgment paradigm: Isolating the pairwise mechanism of the attraction effect. *Proceedings of the 48th Annual Meeting of the Cognitive Science Society.* Oral presentation (140/1511 submissions; 9.3%). [Paper]
- **Rath, T.**, & Marupudi, V. (2025). Re-evaluating the numerical-perceptual distinction in the attraction effect. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society.* [Paper]

Manuscripts Under Review

- **Rath, T.**, Zilka, M., & Shi, D. Gaze Sampling of Motion Relates Differently to Authenticity Judgments for Real and AI-Generated Physics Videos. Full paper under review at the ACM Symposium on Eye Tracking Research & Applications (ETRA 2027). [Draft]
- **Rath, T.**, Srivastava, N., & Srinivasan, N. Unmasking the flaws of triplet-triplet attraction effect measures: Via mathematical analysis and agent-based simulations. Under review at the *Journal of Mathematical Psychology.* [Preprint]

Manuscripts in Preparation

- **Rath, T.** The Proxy Agency Moral Shield: Extended Self, Semantic Fluency, and the Ethics of Agentic AI. Theoretical working manuscript, in final preparation. [Draft]
- **Rath, T.**, Srivastava, N., & Srinivasan, N. Disentangling Ambiguity and Asymmetry in the Attraction Effect: When Comparative Processing Drives Context Dependence. Manuscript in preparation.
- **Rath, T.**, Srivastava, N., & Srinivasan, N. Pairwise comparison architecture in context-dependent choice. Manuscript in preparation.
- Srivastava, C. K., **Rath, T.**, Beuria, J., & Gupta, R. Persistent homology reveals meditation expertise-related regularity in auditory oddball EEG dynamics. In final preparation.

Awards and Honors

- **Cognitive Science Society Student Travel Grant** (CogSci 2026), $1,500 USD; awarded to a first author on a highly rated oral paper.
- **Psychonomic Society Student Travel Award from Developing Nations** (2026), $2,000 USD.
- Ph.D. Institute Assistantship, IIT Kanpur (MoE/MHRD), 2019–2026.

Professional Memberships

Cognitive Science Society; Psychonomic Society; Society for Judgment and Decision Making; Society for Mathematical Psychology.

Research Experience and Contributions

Doctoral Research: Context Effects and Preference Construction

- Designed laboratory and online behavioral experiments in PsychoPy and JavaScript to test how comparison structure and attribute trade-off difficulty shape multi-attribute choice.
- Developed Bayesian and mixed-effects analyses, mathematical arguments, and agent-level simulations to investigate decision mechanisms and the validity of attraction-effect measures under heterogeneous preferences.

Collaborative Human–AI Research: Gaze and Authenticity Judgments

- First-author secondary analysis of 385 real-versus-AI video judgments from 40 participants across 20 matched physics-scene pairs.
- Related motion sampled at fixation to authenticity judgments using same-video, same-time, eccentricity-matched controls and mixed-effects models; compared the information supplied by coupling with conventional gaze summaries and alternative spatial references.

Independent Research: AI Agency and Multi-Agent Systems

- Developing a theoretical framework and experimental program on AI-selected action, felt authorship, and recognition and correction of unintended social consequences.
- Implemented recurrent reinforcement-learning models in PyTorch; developing CARMA to study cross-role value transfer and cooperation in multi-agent systems. Its environment could support future human–AI experiments.

Collaborative Neural-Data Analysis

- Analyzing persistent-homology measures of previously collected, preprocessed auditory-oddball EEG from 54 participants, alongside conventional ERP measures.

Technical Skills

- **Human experiments and measurement:** PsychoPy, JavaScript, OpenSesame; behavioral experiments and psychophysics; EyeLink 1000 Plus.
- **Statistical analysis:** R (lme4, brms, BayesFactor); Python (Pandas, NumPy, SciPy, Statsmodels); Bayesian and mixed-effects modeling; simulation-based evaluation.
- **Computational methods:** Agent-based simulation; PyTorch and recurrent reinforcement learning; multi-agent systems; event-aligned gaze–video analysis; topological data analysis of gaze and preprocessed EEG.

Selected Research Code

- **Situated Agency Alignment / CARMA:** Multi-agent reinforcement-learning environment and candidate cross-role representation intervention for studying cooperation and value transfer. [GitHub]
- **Robust Context Metrics:** Agent-based simulations evaluating attraction-effect measures under heterogeneous preferences. [GitHub]

Teaching and Professional Experience

- **Teaching Assistant, IIT Kanpur:** Human-Centered Computing (CGS616), Jan–Jul 2024; Introduction to Cognitive Science (CGS401), Jul–Dec 2023.
- **Physics Instructor, Chaitanya's Academy and DPS Kalyanpur** (2014–2017): designed curricula and assessments; mentored 500+ students.
- **Senior Software Engineer, Accenture, Pune** (2012–2014): quality-assurance strategy and automated testing for enterprise telecommunications systems.

Selected Talks and Presentations

- **Deliberation eliminates decoys rather than amplifying attraction.** Poster to be presented at the Psychonomic Society Annual Meeting, San Diego (2026).
- **De-obfuscating context effects in decision-making: A simulation study.** Virtual MathPsych/ICCM (2024). [Presentation]
- **Perceptual stimuli with difficult-to-trade-off attributes show a positive attraction effect.** Poster, SJDM, New York (2024).
- **Attribute trade-off difficulty modulates the asymmetric-dominance of the decoy.** Annual Conference of Cognitive Science, IIT Bombay (2024).

References

Prof. Narayanan Srinivasan
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Prof. Nisheeth Srivastava
Department of Computer Science and Engineering, IIT Kanpur
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